



DAFFODIL INTERNATIONAL UNIVERSITY

Department of Computing and Information System (CIS)

Web Development essentials lab

Lab Task-04

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Practice Problem -01:

```
let money=1000;  
let cost=700;  
let result=money-cost;  
  
console.log("The Return is:",result);
```

Output:

```
PS C:\Users\AILAB\OneDrive\Desktop\RAshed> node practice_problem_1.js  
The Return is: 300  
PS C:\Users\AILAB\OneDrive\Desktop\RAshed> █
```

Practice Problem -04:

```
var a=isNaN('11');  
var a=isNaN(2-10);  
  
console.log(a);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\practice_4.js"  
false
```

JS Condition Task-1:

```
let burgerPrice=600;  
if(burgerPrice>500){  
    console.log("Free Coke");  
}  
else{  
    console.log("Coke:30tk");  
}
```

```
}
```

Output:

```
PS C:\Users\AILAB\OneDrive\Desktop\RAshed> node practice_problem_1.js
The Return is: 300
PS C:\Users\AILAB\OneDrive\Desktop\RAshed> |
```

JS Condition Task-2:

```
let weight = 70;
let height = 1.75;
let bmi = weight / (height * height);
console.log("BMI =", bmi);
if (bmi < 18.5) {
    console.log("Underweight");
}
else if (bmi >= 18.5 && bmi <= 24.9) {
    console.log("Normal");
}
else if (bmi >= 25 && bmi <= 29.9) {
    console.log("Overweight");
}
else {
    console.log("Obese");
}
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\BMI_test.js"
BMI = 22.857142857142858
```

JS Condition Task-3:

```
let marks = 85;
if (marks >= 90) {
    console.log("Grade A");
}
else if (marks >= 80) {
    console.log("Grade B");
}
else if (marks >= 70) {
    console.log("Grade C");
}
else if (marks >= 60) {
    console.log("Grade D");
}
else {
    console.log("Grade F");
}
```

Output:

```
PS C:\Users\ATLAB\OneDrive\Desktop\RASHed> node grade_calculator.js
Grade B
PS C:\Users\ATLAB\OneDrive\Desktop\RASHed> █
```

JS Condition Task-4:

```
let myScore = 85;
let friendScore = 65;

if (myScore > 80) {
    if (friendScore > 80) {
        console.log("Go for a lunch");
    }
    else if (friendScore >= 60) {
        console.log("Good luck next time");
    }
    else if (friendScore >= 40) {
        console.log("Keep friend's message unseen");
    }
    else {
        console.log("Block your friend");
    }
}
else {
    console.log("Go home, sleep and act sad");
}
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\nested_ifElse.js"
Good luck next time
```

JS Condition Task-5:

```
let num1=20;
let num2=10;

let result;

if(num1>num2){
    result=num1*2;
}
else{
    result=num1+num2;
}
console.log(result);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\example_5.js"
40

[Done] exited with code=0 in 0.136 seconds
```

JS Condition Task-5: (ternary)

```
let num1 = 20;
let num2 = 15;

let result = (num1 > num2) ? num1 * 2 : num1 + num2;

console.log(result);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\Example_5_ternary.js"
40
```

JS Condition Task-6:

```
let age = 2;
let isStudent = true;
let fare = 800;

if (age < 10) {
    console.log("Free Ticket");
}
else if (isStudent) {
    console.log("Fare =", fare * 0.5);
}
else if (age >= 60) {
    console.log("Fare =", fare * 0.85);
}
else {
    console.log("Fare =", fare);
}
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAsheD\example_6.js"
Free Ticket
```

JS Loop Task-1:

```
let sum=0;
let i=1;
while(true){
    sum+=i;
    if(sum>=100){
        break;
    }
    i++;
}
console.log(sum);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\loop1.js"
105
```

JS Loop Task-2:

```
let arr=[];
for(let i=1; i<=40; i++){
    if(i%2==0){
        arr.push(i);
    }
}
console.log(arr);
console.log(sum);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\loop2.js"
[
  2,  4,  6,  8, 10, 12, 14,
 16, 18, 20, 22, 24, 26, 28,
 30, 32, 34, 36, 38, 40
]
```

JS Loop Task-3:

```
let arr=[];  
for(let i=55; i<=85; i++){  
    if(i%2!=0 && i%5!=0){  
        arr.push(i);  
    }  
}  
console.log(arr);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\Rashed\loop3.js"  
[  
  57, 59, 61, 63, 67,  
  69, 71, 73, 77, 79,  
  81, 83  
]
```

JS Array Task-1:

```
let fruits = ["Apple", "Banana", "Orange", "Mango", "Lichi"];  
console.log(fruits[2]);  
fruits[2] = "Jambura";  
console.log(fruits);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\Rashed\array1.js"  
Orange  
[ 'Apple', 'Banana', 'Jambura', 'Mango', 'Lichi' ]
```

JS Array Task-2:

```
let tourist = ["Cox's Bazar", "Bandarban", "Sylhet"];
tourist.push("Rangamati");
tourist.push("Sajek");
tourist.push("Kuakata");
tourist.pop();
console.log(tourist);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\array2.js"
[ "Cox's Bazar", 'Bandarban', 'Sylhet', 'Rangamati', 'Sajek' ]
```

JS Array Task-3:

```
let books=["Math", "English", "JavaScript", "Physics"];
if (books.includes("JavaScript")) {
    console.log("JavaScript book found");
}
else {
    console.log("Not found");
}
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\array3.js"
JavaScript book found
```

JS Array Task-4:

```
let a = [1,2,3];
let b = "Hello";
let c = 100;

console.log(Array.isArray(a));
console.log(Array.isArray(b));
console.log(Array.isArray(c));
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshe\array4.js"  
true  
false  
false
```

JS Array Task-5:

```
let arr1 = [1,2,3];  
let arr2 = [4,5,6];  
  
let arr3 = arr1.concat(arr2);  
  
console.log(arr1);  
console.log(arr2);  
console.log(arr3);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshe\array5.js"  
[ 1, 2, 3 ]  
[ 4, 5, 6 ]  
[ 1, 2, 3, 4, 5, 6 ]
```

JS String Task-1:

```
let str = "banana";
let count = 0;
for(let ch of str){
    if(ch==='a'){
        count++;
    }
}
console.log(count);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\string1.js"
3
```

JS String Task-2:

```
let str = "bAnanA";
let count = 0;
for(let ch of str){
    if(ch==='a' || ch==='A'){
        count++;
    }
}
console.log(count);
```

Output:

```
[Running] node "c:\Users\AILAB\OneDrive\Desktop\RAshed\string2.js"
3
```

JS String Task-3:

```
const prompt=require("prompt-sync")();
let str=prompt("Enter a string: ");
if(str.includes('a') &&
str.includes('e')&&str.includes('i')&&str.includes('o')&&str.inclu
des('u')){
    console.log("All vawels here");
}
else{
    console.log("Doesn't contaill all vowels");
}
```

Output:

```
> node "I:\Programming\All Coding\Java Script\basic\string3.js"
Enter a string: education
All vawels here
PS I:\Programming\All Coding>
```

JS String Task-4:

```
const prompt=require("prompt-sync")();
let str=prompt("Enter a string: ");

str=str.replaceAll('x','y');
str=str.replaceAll('X','Y');

console.log(str);
```

Output:

```
PS I:\Programming\All Coding> node "i:\Programming\All Coding\Java Script\basic\string4.js"
Enter a string: Xerox xylophone
Yeroy yylophone
PS I:\Programming\All Coding>
```

JS String Task-5:

```
const prompt=require("prompt-sync")();
let str=prompt("Enter a string: ");

let word=str.split(" ");
for(let i=0;i<word.length;i++){
    word[i]=word[i][0].toUpperCase()+word[i].slice(1);
}

let result=word.join(" ");
console.log(result);
```

Output:

```
> node "i:\Programming\All Coding\Java Script\basic\string5.js"
Enter a string: this is rashed
This Is Rashed
PS I:\Programming\All Coding>
```

End